Heat Flow: Gaussian Scaling, Spectral Decay, and Entropy

N-20260621

Why does a localized temperature profile become Gaussian, and why does the heat equation smooth irregular data so quickly? Both questions are answered by the same evolution operator. Conservation fixes the mass of the heat kernel, parabolic scaling fixes its characteristic width, and the Fourier transform explains its damping of high frequencies. When the evolving profile is regarded as a probability density, this same process has an exact entropy-dissipation law.[1]

§1.1 Conservation and Fick's law

For a temperature or concentration $u(x,t)$ on a homogeneous one-dimensional medium, write the flux as $J = -\kappa u_x$, with $\kappa > 0$. Conservation on an interval $[a,b]$ says

$$\frac{d}{dt}\int_a^b u(x,t)\,dx = J(a,t) - J(b,t).$$

Since the interval is arbitrary, the local equation is

$$\boxed{u_t + \partial_x J = 0, \qquad u_t = \kappa u_{xx}.}$$

Thus diffusion is conservation driven by a gradient-dependent constitutive law. At a local maximum $u_{xx} \le 0$, so the temperature there cannot increase through the diffusion term.

Integrating over the whole line, when the flux vanishes at infinity, gives

$$\frac{d}{dt}\int_{\mathbb{R}} u(x,t)\,dx = 0.$$

The profile can spread, but its total mass is fixed. A natural fundamental solution is therefore a family of nonnegative functions of mass one that approaches a point mass as $t \downarrow 0$.

§1.2 The Gaussian follows from parabolic scaling

The equation is invariant under $x \mapsto \lambda x$, $t \mapsto \lambda^2 t$. A mass-one similarity solution should have the form

$$G(x,t) = \frac{1}{2\sqrt{\kappa t}}\Phi(\eta), \qquad \eta = \frac{x}{2\sqrt{\kappa t}}.$$

Substitution into $G_t = \kappa G_{xx}$ yields

$$\Phi'' + 2\eta\Phi' + 2\Phi = 0 \quad \Longleftrightarrow \quad (\Phi' + 2\eta\Phi)' = 0.$$

For a nonnegative integrable profile decaying at both ends, the integration constant must be zero. Hence

$$\Phi' = -2\eta\Phi, \qquad \Phi(\eta) = Ce^{-\eta^2}.$$

[1] MIT 18.152, Lectures 2–5; K. Tang, *Topics in Heat Equations*, §9 (entropy and Fisher information).

The mass normalization $\int_{\mathbb{R}} G\,dx = \int_{\mathbb{R}} \Phi(\eta)\,d\eta = 1$ fixes $C = 1/\sqrt{\pi}$:

$$\boxed{G(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{x^2}{4\kappa t}\right).}$$

Its variance is $2\kappa t$, so the diffusion length scale is $\sqrt{2\kappa t}$. For sufficiently regular and integrable initial data f, the solution on the line is

$$u(x,t) = \int_{\mathbb{R}} G(x-y,t) f(y)\,dy.$$

§1.3 The same Gaussian is a probability transition kernel

For $s, t > 0$, Gaussian convolution obeys

$$G_t * G_s = G_{t+s}.$$

The convolution of two centered normal distributions with variances $2\kappa t$ and $2\kappa s$ has variance $2\kappa(t+s)$; this proves the identity and also explains it probabilistically.

Let B_t be standard Brownian motion, with $B_t \sim N(0,t)$. Then for bounded continuous f,

$$(S_t f)(x) := (G_t * f)(x) = \mathbb{E}\big[f(x + \sqrt{2\kappa} B_t)\big].$$

Brownian motion has independent increments, so

$$S_t S_s f = S_{t+s} f.$$

The operator family S_t is a Markov semigroup, and its infinitesimal generator is $\kappa\partial_x^2$. Indeed, a Taylor expansion under the Gaussian expectation, for smooth f with bounded derivatives, gives

$$\lim_{t \downarrow 0} \frac{S_t f - f}{t} = \kappa f''.$$

Heat diffusion and Brownian motion thus share one Gaussian transition rule: a deterministic PDE describes how the probability distribution of random motion evolves.

§1.4 Fourier modes make smoothing quantitative

Write $\widehat{u}(\xi,t) = \int_{\mathbb{R}} u(x,t) e^{-i\xi x}\,dx$. Fourier transformation of $u_t = \kappa u_{xx}$ gives an ordinary differential equation at each frequency:

$$\partial_t \widehat{u} = -\kappa\xi^2 \widehat{u}, \qquad \boxed{\widehat{u}(\xi,t) = e^{-\kappa\xi^2 t}\widehat{f}(\xi).}$$

The multiplier rapidly suppresses large $|\xi|$. For example, a Fourier mode of frequency 10 decays a hundred times faster in the exponent than one of frequency 1. This is the spectral explanation for instantaneous smoothness at positive times.

The semigroup relation becomes multiplication of multipliers:

$$e^{-\kappa\xi^2 t} e^{-\kappa\xi^2 s} = e^{-\kappa\xi^2 (t+s)}.$$

The Gaussian kernel, Brownian increments and Fourier damping are three exact realizations of the same operation.

§1.5 Boundary Heating and Error Functions

Consider the half-line $x > 0$, initially at temperature zero, with a boundary maintained at unit temperature:

$$u(x,0) = 0 \ (x > 0), \quad u(0,t) = 1 \ (t > 0), \quad \lim_{x\to\infty} u(x,t) = 0.$$

The parabolic scaling suggests $u(x,t) = \theta(\eta)$, where $\eta = x/(2\sqrt{\kappa t})$. Substitution yields

$$\theta'' + 2\eta\theta' = 0.$$

Therefore $\theta'(\eta) = Ce^{-\eta^2}$. The conditions $\theta(0) = 1$, $\theta(\infty) = 0$ give

$$\boxed{u(x,t) = \operatorname{erfc}\left(\frac{x}{2\sqrt{\kappa t}}\right), \qquad \operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}}\int_z^\infty e^{-r^2}\,dr.}$$

The same Gaussian integral that normalized the whole-line kernel now describes the moving transition layer next to a boundary. The profile is exact for $x, t > 0$; its incompatible values at the corner $(0,0)$ reflect the instantaneous change of boundary temperature.

§1.6 Finite intervals have a spectral gap

For $0 < x < L$, impose the fixed-temperature conditions $u(0,t) = u(L,t) = 0$. Separation of variables and orthogonality of sines give

$$u(x,t) = \sum_{n\ge 1} b_n e^{-\kappa(n\pi/L)^2 t}\sin\frac{n\pi x}{L}, \qquad b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx.$$

The lowest spatial eigenvalue is $(\pi/L)^2$. Hence Parseval's identity implies

$$\boxed{\|u(\cdot,t)\|_{L^2(0,L)} \le e^{-\kappa\pi^2 t/L^2}\|f\|_{L^2(0,L)}.}$$

This decay rate can also be obtained without solving explicitly. Multiplying the PDE by u, integrating by parts, and applying the sharp Dirichlet Poincare inequality yields

$$\frac{d}{dt}\frac{1}{2}\int_0^L u^2\,dx = -\kappa\int_0^L u_x^2\,dx \le -\kappa\frac{\pi^2}{L^2}\int_0^L u^2\,dx.$$

Gronwall's inequality reproduces the same exponential estimate. The PDE and its eigenfunction expansion express the same *spectral gap*.

For fixed nonzero endpoint temperatures A, B, first subtract the steady profile

$$h(x) = A + \frac{B-A}{L}x.$$

Then $v = u - h$ obeys homogeneous endpoint conditions and decays toward zero. For insulated endpoints $u_x(0,t) = u_x(L,t) = 0$, the total mass is conserved and the zero Fourier mode survives. The solution approaches its spatial mean rather than zero.

§1.7 Diffusion increases entropy and dissipates information

Now assume $u(x,t) > 0$ is a smooth probability density on $\mathbb{R}$, with sufficient decay to integrate by parts. Its differential Shannon entropy is

$$H(u_t) = -\int_{\mathbb{R}} u(x,t) \log u(x,t)\, dx.$$

Using mass conservation, the heat equation, and integration by parts,

$$\begin{aligned}
\frac{d}{dt} H(u_t) &= -\int_{\mathbb{R}} (\partial_t u)(1 + \log u)\, dx \\
&= -\kappa \int_{\mathbb{R}} u_{xx} \log u\, dx \\
&= \kappa \int_{\mathbb{R}} \frac{u_x^2}{u}\, dx \\
&= \kappa \int_{\mathbb{R}} u\, |\partial_x \log u|^2\, dx.
\end{aligned}$$

The last integral is the *Fisher information* $I(u_t)$, so

$$\boxed{\frac{d}{dt} H(u_t) = \kappa I(u_t) \geq 0.}$$

Entropy increases because the density becomes more spread out; the rate is precisely its Fisher information. This identity is a concrete bridge between heat flow and information geometry.

For the Gaussian kernel $G_t = N(0, 2\kappa t)$,

$$H(G_t) = \frac{1}{2} \log(4\pi e \kappa t), \qquad I(G_t) = \frac{1}{2\kappa t}.$$

Direct differentiation gives $H'(G_t) = 1/(2t) = \kappa I(G_t)$, confirming the general identity on the fundamental solution.

The wave equation $w_{tt} = c^2 w_{xx}$ behaves differently. Its Fourier modes oscillate instead of acquiring the damping factor $e^{-\kappa \xi^2 t}$; appropriate wave energy is conserved instead of dissipated. This difference between first- and second-order time evolution explains why heat flow smooths and loses detail while wave motion preserves oscillatory information.